Multimode Raman Baselines and Cost Choice

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-28

Abstract

This note summarizes the current multimode-fiber Raman-suppression lane. The accepted result is a qualified simulation finding: in an idealized six-mode GRIN-50 model, one shared spectral phase reduced the simulated Raman-band fraction by 23.9 dB on the final $N_t = 8192$ validation grid while passing corrected temporal-edge diagnostics. A lower-resolution regularized candidate reached 31.7 dB, but the conservative headline is the high-resolution validation. The most important part of the story is that an earlier large-gain solution was rejected after raw edge diagnostics exposed temporal-window contamination. The current result is therefore meaningful, but not a generic experimental claim about arbitrary multimode fibers.

Claim Status

Status: qualified simulation result
The MMF lane is closed/exploring. It is usable as qualified simulation evidence with clear claim boundaries: shared spectral phase shaping suppresses Raman-band generation in one idealized GRIN-50 setting under strict diagnostics. It is not yet a paper-grade or experiment-grade universal MMF result.

1 Question and Thesis

The single-mode notes optimize one spectral phase for one propagation mode. In a multimode fiber, the field has several mode amplitudes:

$$\hat{u}_{m,k}(z),$$

where m indexes mode and k indexes frequency. The question is:

Can one shared spectral phase reduce Raman generation across a multimode field?

The current evidence says yes in a constrained simulation. The accepted high-resolution case is GRIN-50 with six scalar modes, length $L = 2\,\mathrm{m}$, power $P = 0.20\,\mathrm{W}$, $N_t = 8192$, and a 96 ps time window. The optimized shared phase reduces the raw summed Raman-band metric from $-17.37\,\mathrm{dB}$ to $-41.25\,\mathrm{dB}$.

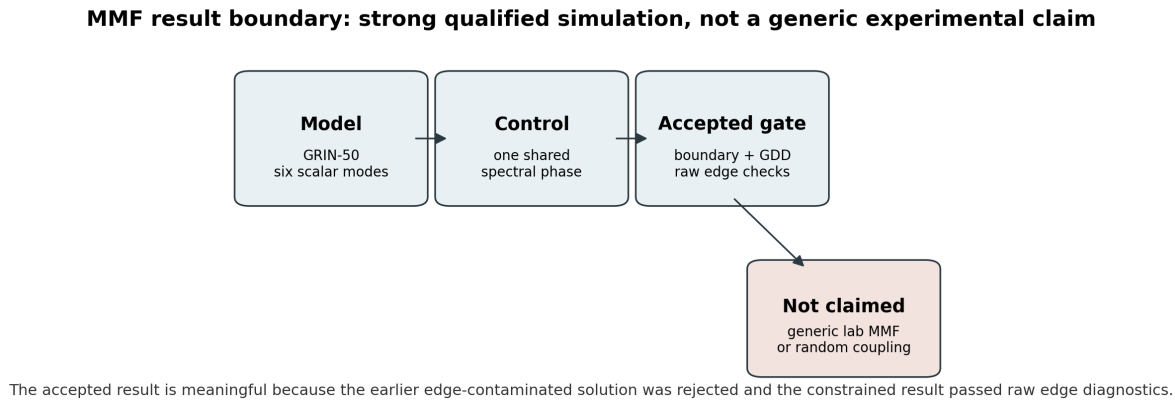


Figure 1: The current MMF result is intentionally narrow. The project has a credible constrained simulation candidate, not a generic experimental claim for all multimode fibers or all launch conditions.

2 Why MMF Is Subtle

Multimode fibers are not just single-mode fibers with a bigger vector. Raman energy can appear differently by mode, launch composition matters, and mode coupling can change the nonlinear evolution. Experiments and simulations of graded-index multimode fibers show mode-resolved Raman cascades [1], nonlinear beam self-cleaning [2], and launch/wavefront sensitivity [3, 4].

That literature shapes the claim boundary. A summed output spectrum is not enough. A credible MMF result needs:

- a clear objective: summed, fundamental-mode, or worst-mode Raman fraction;
- mode-resolved inspection, not only a total spectrum;
- temporal-window diagnostics, because pseudospectral propagation uses periodic time windows [5];
- an explicit statement of launch and coupling assumptions.

TL;DR

The MMF result is harder to explain because there are two kinds of structure: frequency structure and mode structure. The shared phase only controls the frequency structure. It does not independently control the transverse modal wavefront, and it does not prove robustness to every possible launch.

3 Model, Objective, and Gradients

The accepted model is:

$$\text{GRIN-50}, \quad M = 6, \quad L = 2\,\text{m}, \quad P = 0.20\,\text{W}.$$

The same spectral phase is applied to all modes:

$$\hat{u}_{m,k}(0;\phi) = \hat{u}_{m,k,0} e^{i\phi_k}.$$

The primary objective is the mode-summed Raman fraction:

$$J_{\text{sum}} = \frac{\sum_m \sum_{k \in \mathcal{B}_{\text{Raman}}} |\hat{u}_{m,k}(L)|^2}{\sum_m \sum_k |\hat{u}_{m,k}(L)|^2}.$$

The diagnostic report also computes a fundamental-mode fraction and worst-mode fraction so the improvement cannot hide inside a summed detector. For the accepted run:

$$J_{\text{sum}} = -49.69\,\text{dB}, \qquad J_{\text{fund}} = -49.65\,\text{dB}, \qquad J_{\text{worst}} = -45.35\,\text{dB}.$$

The shared-phase gradient sums the modal sensitivities:

$$\frac{\partial J}{\partial \phi_k} = 2\,\text{Re} \sum_m \lambda_{m,k}(0)^*\, i\, \hat{u}_{m,k}(0).$$

This is the same phase-chain rule as the single-mode case, except every mode contributes to the same phase pixel.

Where the Shared-Phase Gradient Comes From

Start with one frequency sample k. The optimizer is not choosing a separate phase for every mode. It chooses one number ϕ_k, and that same number is applied to all modal spectra:

$$\hat{u}_{m,k}(0;\phi) = \hat{u}_{m,k,0} e^{i\phi_k}.$$

A small phase change therefore gives

$$\delta \hat{u}_{m,k}(0) = i \hat{u}_{m,k}(0) \delta\phi_k.$$

If $\lambda_{m,k}(0) = \partial J / \partial \hat{u}_{m,k}(0)^*$ is the adjoint sensitivity at the input, then the real objective changes by

$$\delta J = 2\,\text{Re} \sum_{m,k} \lambda_{m,k}(0)^*\, \delta \hat{u}_{m,k}(0).$$

Substituting the phase perturbation and collecting the coefficient of $\delta\phi_k$ gives the formula above. The only new MMF feature is the modal sum: all modes vote on the same phase knob.

TL;DR for the Board

Think of each wavelength as one knob shared by six mode channels. Turning that knob rotates the complex field in every mode at that wavelength. The adjoint tells us whether each modal rotation helps or hurts the Raman objective, and the gradient adds those six votes into one update for the shared knob.

Accepted Regularized Surface

The accepted constrained run uses a dB-scaled surface of the form

$$10 \log_{10} \left(J_{\text{sum}} + \lambda_{\text{gdd}} R_{\text{gdd}} + \lambda_{\text{boundary}} R_{\text{boundary}} \right),$$

with

$$\lambda_{\text{gdd}} = 10^{-4}, \qquad \lambda_{\text{boundary}} = 0.05.$$

The boundary term is central to this note because it distinguishes the rejected result from the accepted one.

4 Diagnostic Correction

The earlier MMF result looked strong, but it was not accepted. The problem was temporal-window trust. The corrected convention is:

$$\hat{u} = \text{ifft}(u_t), \qquad u_t = \text{fft}(\hat{u}).$$

After this correction, the unregularized optimized output still had about 5.02×10^{-2} temporal-edge energy. That is too high, so the result was rejected as a boundary artifact.

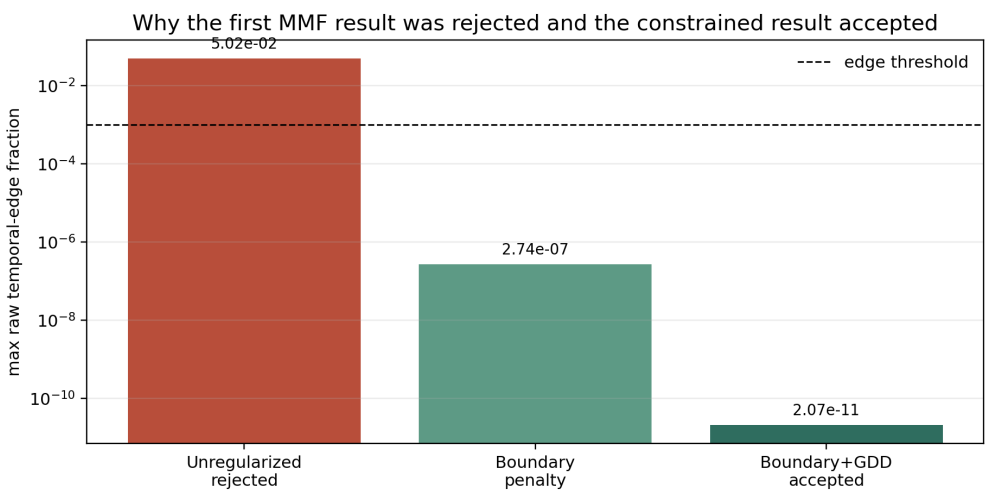


Figure 2: The unregularized solution is rejected because its raw edge fraction is too high. Boundary and GDD regularization recover strong suppression while driving the raw edge fraction far below the trust threshold.

TL;DR for the Trust Gate

If a pulse reaches the edge of the simulated time window, the FFT can wrap it around as if time were periodic. A result can then look good for the wrong reason. The accepted MMF result is important because it survives the raw-edge check, not just because it has a low Raman number.

5 Implementation Surface

Provenance and Reproducibility Check

The accepted numbers in this note come from the dedicated MMF window-validation driver, not from a presentation-only plotting script. The run wrote the standard image set, the mode-resolved spectra, and the validation summary. It was then visually inspected because the core risk in this lane is numerical trust, not just whether an optimizer prints a lower objective.

The accepted high-resolution run uses $N_t = 8192$, 96 ps, six scalar modes, and the same boundary+GDD penalties as the earlier accepted $N_t = 4096$ candidate. It stopped through the driver-side time limit, but the patched optimizer returned the best observed phase, wrote the standard image set, and passed raw-edge diagnostics. This is why the claim is now stronger than the earlier grid-limited result, while still remaining a qualified simulation finding rather than a broad MMF generalization.

Code Path

Component	Role
MMF setup	Builds the GRIN-50 preset, mode launch, grid, fiber, and Raman band.
Shared-phase optimizer	Computes the MMF forward solve, adjoint solve, summed objective, and shared phase gradient.
Window-validation driver	Reruns the threshold MMF case with larger windows and regularizers, then writes the validation summary.
Cost helpers	Provide summed, fundamental-mode, and worst-mode Raman fractions.
Standard-image helper	Saves phase profile, phase diagnostic, optimized heat map, and no-shaping heat map.
Front-layer config	Supports planning and dry-run validation only; real MMF propagation remains burst/heavy-compute work.

6 Experimental Ladder

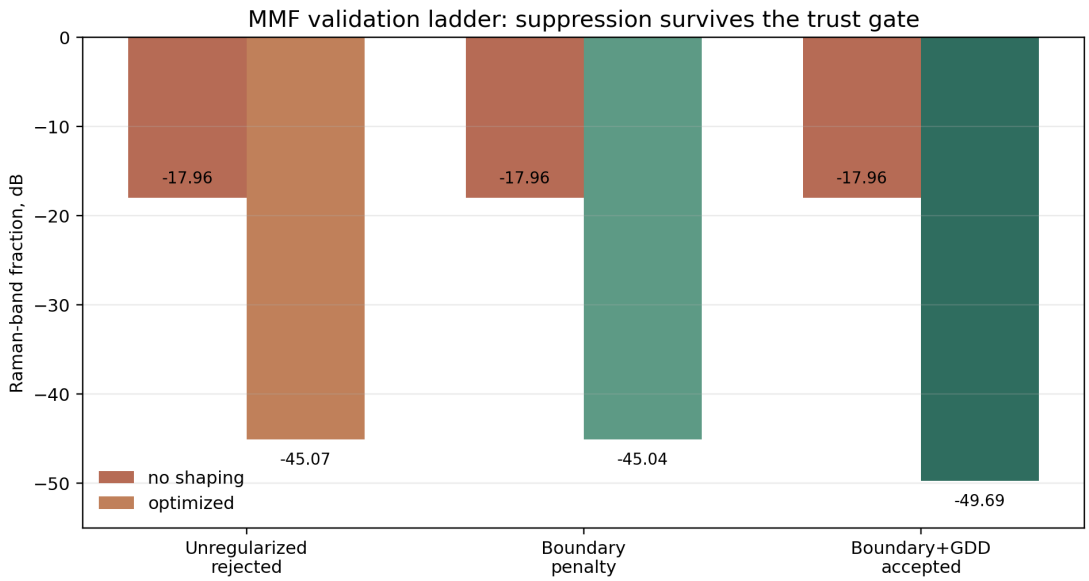


Figure 3: MMF validation ladder. The final accepted result is not just the deepest-looking run; it is the deepest run that also passes the raw temporal edge diagnostics among the tested candidates.

Case	**Objective dB**	**Improvement**	**Max edge**	**Verdict**
Unregularized	-45.07	27.12 dB	5.02×10^{-2}	reject
Boundary penalty	-45.04	27.09 dB	2.74×10^{-7}	pass
Boundary+GDD	-49.69	31.73 dB	2.07×10^{-11}	accepted
High-res boundary+GDD	-41.25	23.88 dB	3.59×10^{-13}	accepted

Table 1: All rows use $L = 2\,\mathrm{m}$, $P = 0.20\,\mathrm{W}$, and 96 ps. The first three rows use $N_t = 4096$; the high-resolution row uses $N_t = 8192$. The unregularized row is scientifically useful because it explains what had to be fixed, but it is not accepted as a valid result.

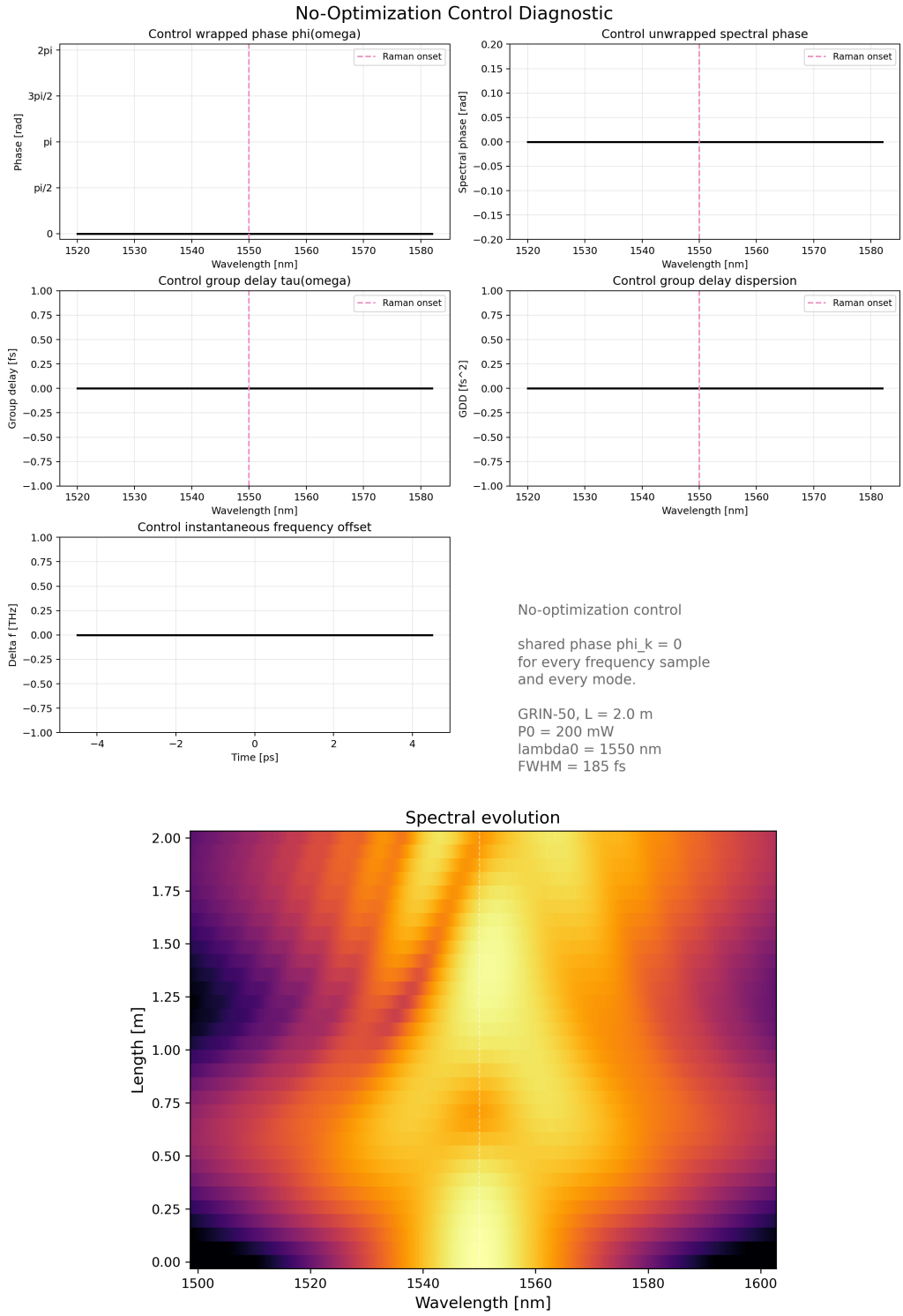


Figure 4: No-optimization control for the accepted MMF operating point. The top panel records the flat shared phase $\phi_k = 0$; the bottom panel is the actual no-shaping spectral-evolution heat map used as the control reference.

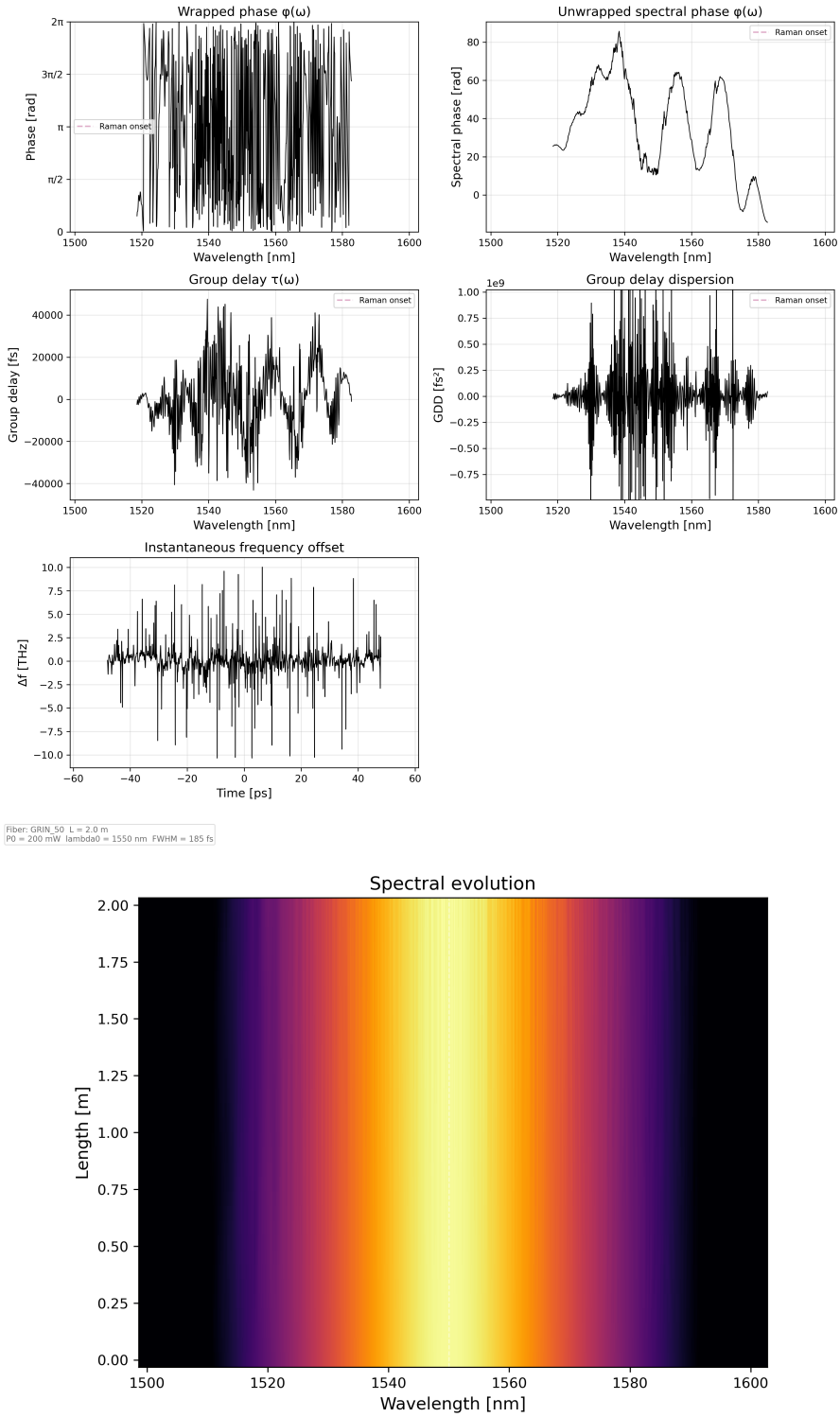
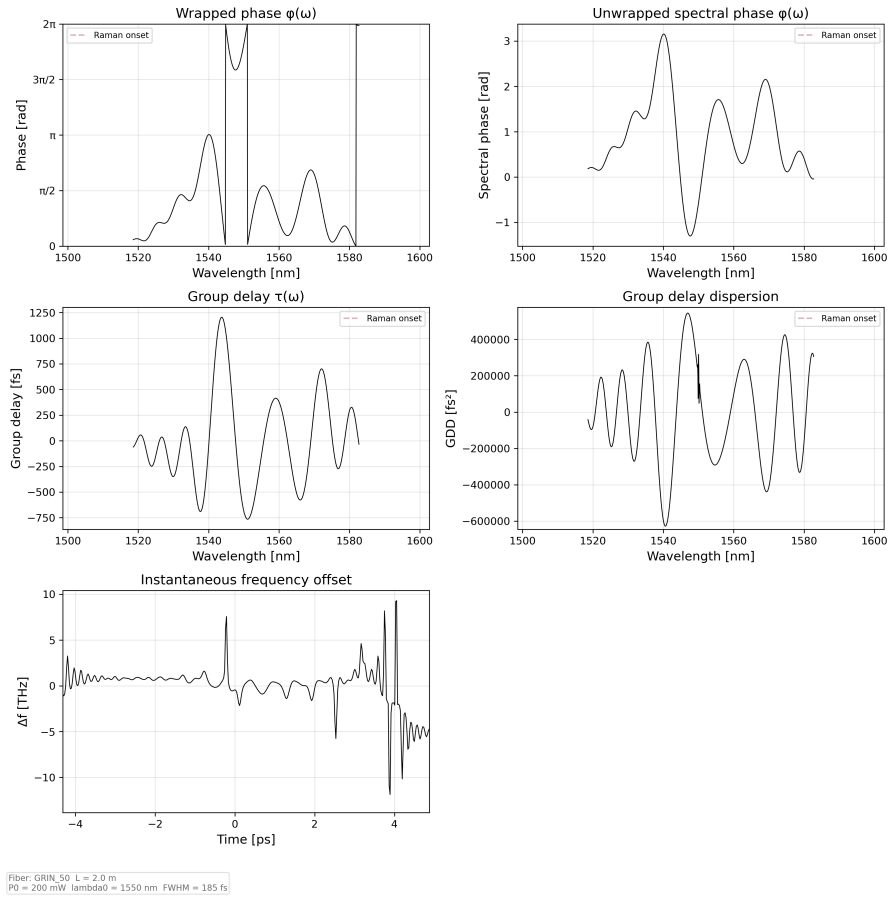


Figure 5: The unregularized run visually suppresses Raman output, but it is not accepted because corrected raw-edge diagnostics show temporal-window contamination.



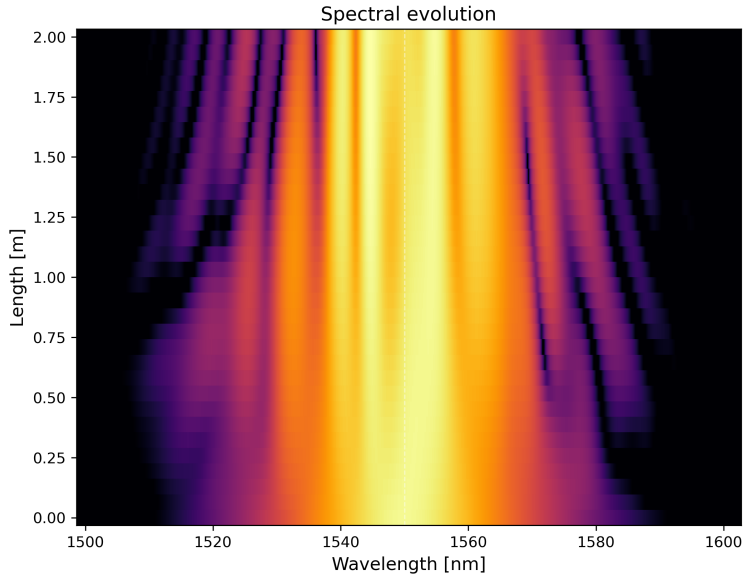
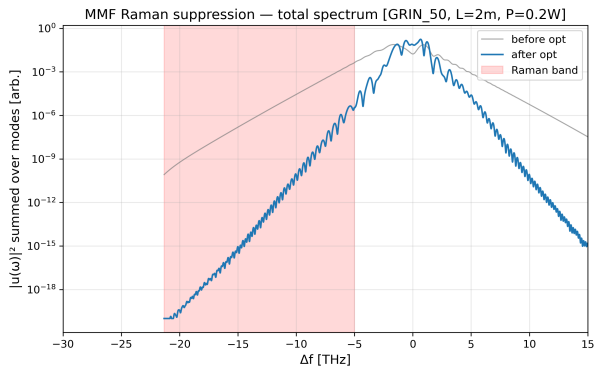
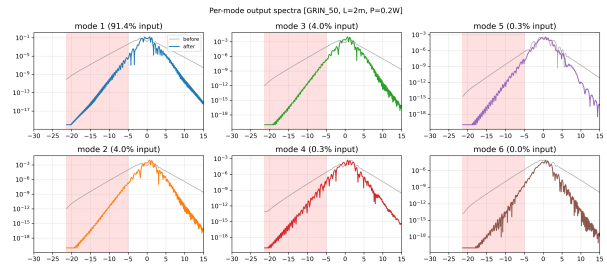


Figure 6: Accepted constrained MMF result: the original $N_t = 4096$ candidate reached $-17.96 \rightarrow -49.69\,\mathrm{dB}$; the final $N_t = 8192$ refinement reached $-17.37 \rightarrow -41.25\,\mathrm{dB}$ with raw edge fractions below 4×10^{-13}.

7 Representative Result Pages

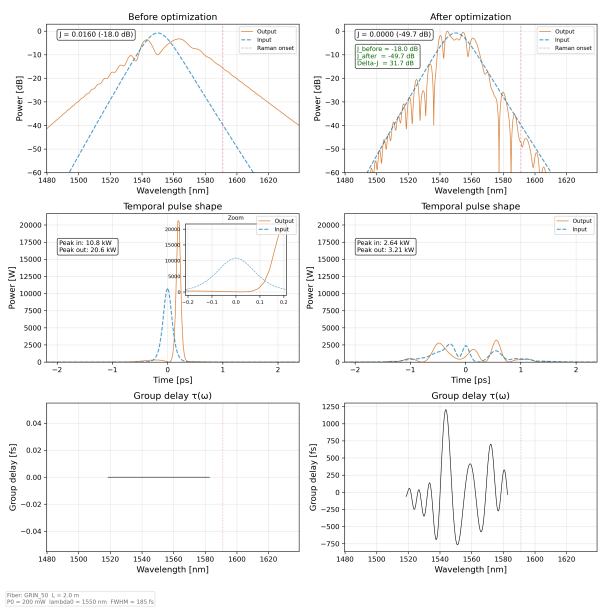


Total spectrum.

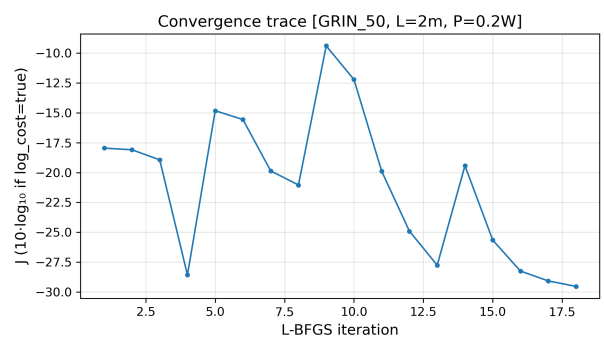


Per-mode spectrum.

Figure 7: Mode-resolved inspection matters. The accepted run is not only a summed-detector accounting win; the launched modes also show suppression.



Accepted phase profile.



Convergence trace.

Figure 8: The accepted phase is still engineered and oscillatory. This is a simulation candidate, not a lab actuator prescription.

8 Interpretation

The MMF lane changed from "blocked by invalid-window" to "qualified simulation result." The key scientific behavior survives after the trust correction: boundary and GDD penalties can keep the temporal window clean while still producing large Raman suppression.

The result should be presented with discipline. It supports shared spectral phase as a possible control mechanism in an idealized GRIN-50 MMF model. It does not yet prove robustness to launch changes, random mode coupling, more modes, or experimental wavefront shapers.

9 Presentation Capsule

If this note becomes a presentation block, the clean message is:

- MMF is harder because Raman is mode-resolved and launch/coupling assumptions matter.
- The first large-gain result was rejected because the pulse touched the temporal window boundary.
- The accepted constrained run still gives a 23.9 dB reduction at $N_t = 8192$ and passes raw edge diagnostics.
- The honest claim is an idealized six-mode GRIN-50 simulation result, not generic experimental MMF Raman suppression.

10 Limitations

- **No broad experimental claim.** The model is idealized and uses six scalar modes.
- **Shared spectral phase only.** This is not independent spatial wavefront control of each mode.
- **Launch sensitivity open.** Default launch is LP01-heavy; other launch compositions still need a matrix of checks.
- **Random coupling open.** The accepted result does not include a random or degenerate mode-coupling robustness sweep.
- **Grid-refinement gate closed for this lane.** The $N_t = 8192$, 96 ps refinement completed with standard images and clean raw edge diagnostics.
- **Phase realism open.** The accepted phase has sub-ps to roughly ps-scale group-delay structure and should be simplified before lab claims.

11 Reproduction Capsule

Planning-only front-layer check:

```
julia -t auto --project=. scripts/canonical/run_experiment.jl \
  --dry-run grin50_mmf_phase_sum_poc
```

Accepted constrained candidate, to run on burst or equivalent heavy compute:

```
MMF_VALIDATION_SAVE_DIR=results/raman/mmf_window_validation_gdd_nt8192_final \
MMF_VALIDATION_CASES=threshold \
MMF_VALIDATION_MAX_ITER=4 \
```

```
MMF_VALIDATION_THRESHOLD_TW=96 \
MMF_VALIDATION_THRESHOLD_NT=8192 \
MMF_VALIDATION_LAMBDA_BOUNDARY=0.05 \
MMF_VALIDATION_LAMBDA_GDD=1e-4 \
MMF_VALIDATION_F_CALLS_LIMIT=80 \
MMF_VALIDATION_TIME_LIMIT_SECONDS=10800 \
julia -t auto --project=. scripts/research/mmf/mmf_window_validation.jl
```

Expected closeout:

- validation summary table;
- phase profile and phase diagnostic;
- optimized and no-shaping heat maps;
- total and per-mode spectra;
- raw input/output temporal-edge fractions;
- explicit statement of whether the result is accepted, rejected, or compute-blocked.

12 Sources

References

[1] H. Pourbeyram, G. P. Agrawal, and A. Mafi. "Stimulated Raman scattering cascade spanning the wavelength range of 523 to 1750 nm using a graded-index multimode optical fiber." https://arxiv.org/abs/1301.6203

[2] K. Krupa et al. "Spatial beam self-cleaning in multimode fibres." *Nature Photonics*, 2017. https://doi.org/10.1038/nphoton.2017.32

[3] E. Deliancourt et al. "Wavefront shaping for optimized many-mode Kerr beam self-cleaning in graded-index multimode fiber." https://arxiv.org/abs/1902.04453

[4] O. Sidelnikov et al. "Effect of random mode coupling on Kerr beam self-cleaning in graded-index multimode fibers." https://arxiv.org/abs/1908.07745

[5] O. Melchert and A. Demircan. "A python package for ultrashort optical pulse propagation in terms of forward models for the analytic signal." https://arxiv.org/abs/2104.11649